

Shouwei Gao

Ph.D. Candidate in Computer Science
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Corvallis, OR, USA

RESEARCH INTERESTS LLM Serving & Inference Systems · Efficient AI & Model Compression · AI for Science (SciML) · Explainable AI

EDUCATION **Oregon State University** Aug. 2023 – Present
Ph.D. in Computer Science — Corvallis, OR
University of Electronic Science and Technology of China Sep. 2020 – Jun. 2023
M.Eng. in Electronic Information — Chengdu, China
Xihua University Sep. 2014 – Jun. 2018
B.S. in Energy and Power Engineering — Chengdu, China

PUBLICATIONS **Flying Serving: On-the-Fly Parallelism Switching for Large Language Model Serving.** Shouwei Gao, Junqi Yin, Feiyi Wang, Wenqian Dong. *ACM International Conference on Supercomputing* (ICS 2026). **(Best Paper Candidate)**
LUMOS: Democratizing SciML Workflows with L_0 -Regularized Learning for Unified Feature and Parameter Adaptation. Shouwei Gao, Xu Zheng, Dongsheng Luo, Sheng Di, Wenqian Dong. *IEEE International Parallel and Distributed Processing Symposium* (IPDPS 2026).
Evaluating LLM Coding Agents on SZ-Family Lossy Compression Across Architectures. Changqing Li, Shouwei Gao, Kai Zhao, Sheng Di, Wenqian Dong. *IEEE International Parallel and Distributed Processing Symposium, HPAI4S Workshop* (IPDPS-W 2026).
HurriCast: Synthetic Tropical Cyclone Track Generation for Hurricane Forecasting. Shouwei Gao, Meiyan Gao, Yuepeng Li, Wenqian Dong. *AAAI Spring Symposium Series, GenAI@Edge* (SSS-25, 2025).
Eddy Current Array for Defect Detection in Finely Grooved Structure Using MSTSA Network. Shouwei Gao, Yali Zheng, Shengping Li, Jie Zhang, Libing Bai, Yaoyu Ding. *Sensors* (MDPI, 2024).
Eddy Current Array for Defects Detection Based on Spatiotemporal Self-Attention Network. Shouwei Gao, Yali Zheng, Jie Zhang, Libing Bai. *International Conference on Sensing, Measurement & Data Analytics* (ICSMD 2022).

PREPRINTS & MANUSCRIPTS UNDER REVIEW **DOLMA: A Data Object Level Memory Disaggregation Framework for HPC Applications.** Haoyu Zheng, Shouwei Gao, Jie Ren, Wenqian Dong. *arXiv preprint* (arXiv:2512.02300, 2025).
PAC Learnability under Explanation-Preserving Graph Perturbations. Xu Zheng, Farhad Shirani, Tianchun Wang, Shouwei Gao, Wenqian Dong, Wei Cheng, Dongsheng Luo. *arXiv preprint* (arXiv:2402.05039, 2024).
Elastic Parallelism for LLM Serving. Shouwei Gao, et al. *European Conference on Computer Systems* (EuroSys 2027). Under review.
Efficient AI Foundation Models. Shouwei Gao, et al. *International Conference for High Performance Computing, Networking, Storage, and Analysis* (SC 2026). Under review.

TALKS &
PRESENTATIONS

- **Oak Ridge National Laboratory** — “Flying Serving: On-the-Fly Parallelism Switching for Large Language Model Serving.” 2025
- **IPDPS 2026** — “LUMOS: Democratizing SciML Workflows with L_0 -Regularized Learning for Unified Feature and Parameter Adaptation.” 2026
- **AAAI 2025, GenAI@Edge Symposium** — Paper presentation, “HurriCast.” 2025

TEACHING

Graduate Teaching Assistant, Oregon State University

- CS 475/575 — Introduction to Parallel Programming Spring 2026
- CS 599 — Advanced Topics in Deep Learning & High-Performance Computing
- CS 381 — Programming Languages
- Operating Systems

HONORS & AWARDS

- **OSU Graduate Student Scholarly Presentation Award**, Oregon State University 2026
- **Best Paper Candidate**, ACM International Conference on Supercomputing (ICS) 2026
- Second Prize (7th / 611), Huawei Software Elite Challenge 2022
- Huawei Embedded Software Contest (8th / 48) 2022
- ZTE Software Elite Challenge — Image Denoising Algorithm Contest 2022

RESEARCH &
WORK EXPERIENCE

Research Intern, Oak Ridge National Laboratory Sep. 2025 – Present
Large-scale LLM deployment and serving on multi-GPU clusters.

Graduate Research Assistant, Oregon State University Aug. 2023 – Present
LLM serving systems, efficient AI, and AI for science.

Automation Equipment Maintenance Engineer Jul. 2018 – Aug. 2019
Sichuan Datang International Ganzi Hydropower Development Co., Ltd.

TECHNICAL SKILLS

Programming: Python, C/C++, MATLAB, Java
ML & Systems: PyTorch, vLLM, SGLang, TensorFlow, CUDA programming
Profiling & Tools: NVIDIA Nsight Systems, NVIDIA Nsight Compute, Qt, Git